Knee Exoskeleton — Bill of Materials

Unilateral build · Generated Nov 3, 2026 · Prices in USD, excluding tax, customs and shipping

Mechanical	Electrical	Low-Level Specific PCB	High-Level Specific PCB	Grand Total
$275.38	**$753.13**	**$115.78**	**$456.24**	**$1,600.53**

Build notes

- Items in this table are for a unilateral purchase. For bilateral: +1 motor, +2 IMUs, +1 Teensy.
- Most purchased items are bulk, so there should be enough to cover a bilateral system too. Modify count and purchase per your needs.
- Filament: the whole bilateral system can be printed with 1.5–2 kg of filament. Order more as needed.
- Custom PCBs: upload the Gerber files to JLCPCB. Prices exclude sales tax, customs tax and shipping.

Mechanical						$275.38	
	Name	Part Name	Provider	Unit Price	Qty	Total	Link (where to buy)
☐	**M4 Fastener Kit**	Mywish 380 Pcs M4 Screw Nuts and Bolts Assortment Kit, 304 Stainless Steel Button Head Socket Cap Metric Screws Bolts, Spring & Flat Lock Washers/Nuts/Hex Wrenches, M4*8/10/12/16/20/25/30	Amazon	$8.39	1	$8.39	a.co/d/0dEzoNLS
☐	**Standoff Kit**	HELIFOUNER 242 Pieces M2.5 Male Female Hex Brass Spacers Standoffs Screws Nuts Assortment Kit, Threaded Pillar for Printed Circuit Board Motherboard	Amazon	$14.69	1	$14.69	a.co/d/0955o79O
☐	**PAHT-CF Filament**	PAHT-CF, 1kg	Bambu	$94.99	2	$189.98	us.store.bambulab.com/products/paht-cf
☐	**M3 Motor Fastener**	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw 90 Degree Countersink, M3 x 0.5 mm Thread Size, 10 mm Long	McMaster	$6.36	1	$6.36	mcmaster.com/91294A130
☐	**Foam**	HOMBYS Upholstery Foam Memory, 1/2" Height 72" L x 24" W High Density Foam Padding Couch Cushion Support,Cut to Size Foam Furniture Replacement Pad for Sofa, Mattress Sag Support, 0.5" H, Blue	Amazon	$17.99	1	$17.99	a.co/d/09KuAOxC
☐	**3D mesh**	Tong Gu 3D Air Mesh Fabric Net Cloth Three Layer Sandwich Spacer 3mm Thick Craft Seat Sew Material (Black)	Amazon	$19.99	1	$19.99	a.co/d/0cibKLjl
☐	**Nylon Fabric**	Precut Nylon Fabric (2 Yards x 57 inch) - Water Repellent Ripstop Fabric, Lightweight, Durable - Used for Covers, Flags, Sacks, Bags, Kites, Outdoor Crafts and DIY - Black	Amazon	$11.99	1	$11.99	a.co/d/0bUM3wLF
☐	**Belt**	Men's Belt,Nylon Belts for Men with Plastic Buckle,Quick Release Tactical Belt,Sliding Ring Buckle	Amazon	$5.99	1	$5.99	a.co/d/03hOzpND

Electrical						$753.13
	Name	Part Name	Provider	Unit Price	Qty	Total	Link (where to buy)
☐	**AK80-9 V3 Motor**	AK80-9 V3.0 KV100	CubeMars	$579.90	1	$579.90	cubemars.com/product/ak80-9-v3-0-rob…
☐	**IMU**	Adafruit TDK InvenSense ICM-20948 9-DoF IMU	Adafruit	$19.95	2	$39.90	adafruit.com/product/4554
☐	**Buck Converter New**	EBOOT 3 Pack MP1584EN DC-DC Buck Converter 24V to 12V 9V 5V 3V Adjustable	Amazon	$7.65	1	$7.65	a.co/d/0db9gjQJ
☐	**Buck Converter Old**	5 Pack LM2596 DC to DC Buck Converter 3.0-40V to 1.5-35V Adjustable Voltage Regulator Electronic Voltage Stabilizer Power Supply Step Down Module	Amazon	$7.99	1	$7.99	a.co/d/0cBbuDSb
☐	**Battery**	HRB 2packs 6S 22.2V 3300mAh 60C XT60 RC Lipo Battery	HRB	$75.99	1	$75.99	hrb-power.com/products/hrb-2packs-6s-…
☐	**Emergency Stop (E-STOP)**	APIELE 19mm Metal Emergency Stop Push Button Swicth Latching 12-250V Stianless Steel 0.74" Mouting Hole	Amazon	$12.99	1	$12.99	a.co/d/0gLPYfHd
☐	**Shank IMU board**	Solderable Breadboard Mini PCB Board Gold-Plated Solder Breadboard for DIY Electronics Projects Apply to Arduino Soldering Projects 2.0"x1.5" Mixed Color(Pack of 6 Pcs)	Amazon	$7.99	1	$7.99	a.co/d/0hml2zFF
☐	**2 pin JST - GH Connector**	JST SM02B-GHS-TB(LF)(SN)	LCSC	$0.3728	10	$3.73	lcsc.com/product-detail/C189893.…
☐	**2 pin JST - GH Cable**	20 PCS 2 PIN JST GH Connectors on Both Sides with Electric Wire 1007 28 AWG(Total Length:200MM)	Amazon	$16.99	1	$16.99	a.co/d/07Uxlu8K

Low-Level Specific PCB						$115.78
	Name	Part Name	Provider	Unit Price	Qty	Total	Link (where to buy)
☐	**Teensy 4.1**	Teensy 4.1 (Headers)	Sparkfun	$35.95	1	$35.95	sparkfun.com/teensy-4-1-headers.html
☐	**Shank IMU Board**	Solderable Breadboard Mini PCB Board Gold-Plated Solder Breadboard for DIY Electronics Projects Apply to Arduino Soldering Projects 2.0"x1.5" Mixed Color(Pack of 6 Pcs)	Amazon	$7.99	1	$7.99	a.co/d/065fXxdn
☐	**Shank IMU Wiring**	TUOFENG 22 AWG Solid Wire Kit, 6 Colors, 30 ft Each	Amazon	$16.19	1	$16.19	a.co/d/08nN9mnW
☐	**Shank IMU cable**	Cable Matters 10Gbps 5-Pack Snagless Cat 6 Ethernet Cable, 2ft, Black	Amazon	$11.99	1	$11.99	a.co/d/06NZkQ9e
☐	**CAN Tranciever Chip**	TJA1051T/3/1J	DijiKey	$1.74	2	$3.48	digikey.com/en/products/detail/nxp-…
☐	**4.7 uF Cap**	CS1206KKX7R0BB105	DijiKey	$0.28	10	$2.80	digikey.com/en/products/detail/yage…
☐	**60 ohm SMD Res**	RMCF1206FT60R4	DijiKey	$0.01	25	$0.25	digikey.com/en/products/detail/stac…

Low-Level Specific PCB						$115.78	
	Name	Part Name	Provider	Unit Price	Qty	Total	Link (where to buy)
☐	**XT30PW Connection**	Connector XT30PW30-M.G.* Male Plug Right Angle for RC ESC	LCSC	$0.38	3	$1.14	lcsc.com/product-detail/C431092....
☐	**RJ45 Header**	Treedix 4 Pcs RJ45 8-pin Connector (8P8C) and Breakout Board Kit Compatible with Ethernet DMX-512 RS-485 RS-422 RS-232 (Unassambled)	Amazon	$8.99	1	$8.99	a.co/d/0dUZjbGj
☐	**Knee PCB V1**	Custom PCB, fabricate from Gerber files	JLCPCB	$27.00	1	$27.00	Gerber files (JLCPCB)

High-Level Specific PCB						$456.24	
	Name	Part Name	Provider	Unit Price	Qty	Total	Link (where to buy)
☐	**CAN PAL**	Adafruit CAN Pal - CAN Bus Transceiver - TJA1051T/3	Adafruit	$3.95	1	$3.95	adafruit.com/product/5708
☐	**Jetson**	Jetson Orin Nano Super Developer Kit	Nvidia	$399.00	1	$399.00	marketplace.nvidia.com/en-us/enterprise/roboti...
☐	**Fuse Holder**	XFCN PTF-15	LCSC	$0.288	5	$1.44	lcsc.com/product-detail/C842712....
☐	**Fuse**	BOJACK 15 Values 200 pcs Fast-Blow Glass Fuses Assortment Kit 5x20mm 250V 1 5 10 15 20A 6x30mm 250V 1 2 3 5 6 7 8 10 15 20A amp packag in a Clear Plastic Box	Amazon	$9.99	1	$9.99	a.co/d/0aJ6Ptv4
☐	**XT30 2+2 Upright**	PCB Connector XT30PB(2+2)-M.G.* Male Plug Power + Signal Connector Vertical PCB Mount Gold Plated Black	LCSC	$0.4423	4	$1.77	lcsc.com/product-detail/C1926802...
☐	**XT60 Battery Connection**	XT60Connector XT60-M. 2-Pin Male Plug Gold Plated for LiPo Battery	LCSC	$0.354	5	$1.77	lcsc.com/product-detail/C98733.h...
☐	**XT30 E-Stop Connection**	PCB Vertical Connector XT30UPB30-M.G.* Male Plug for Battery Board	LCSC	$0.2664	5	$1.33	lcsc.com/product-detail/C428721....
☐	**Jetson Header Extension**	Frienda 8 Pcs 2 x 20 (40 Pin) Extra Tall Female	Amazon	$9.99	1	$9.99	a.co/d/05zeNa1Y
☐	**Jetson V4 PCB**	Custom PCB, fabricate from Gerber files	JLCPCB	$27.00	1	$27.00	Gerber files (JLCPCB)